

# Gene Pulser® Electroprotocol

**Cell Type** Bacterial, gram negative

**Molecules Electroporated** DNA: plasmid, covalently closed, circular, 13 kB

**Species Used** Cyanobacteria, primarily filamentous, *Anabaena* spp.

## Before the Pulse

**Cell Growth Medium** Growth medium of Allen and Arnon, diluted 8-fold (AA/8) [see notes]

**Growth Phase at Harvest** O.D. (600) = not given mid-log

**Pre-pulse Incubation** none

**Wash Solution** 1.0 mM HEPES, pH 7.4

## The Pulse

**Instruments Used** Gene Pulser® apparatus

**Electroporation Temperature** 4 °C

**Electroporation Medium** 1.0 mM HEPES pH 7.4

**Cuvette Gap** 0.2 cm

**Cell Density** 5 x 10<sup>8</sup> (8) cells / ml

**Voltage** 1.6 kV

**Volume of Cells** 40 µl

**Field Strength** 8 kV/cm

**DNA Concentration** 100 µg / ml

**DNA Resuspension Buffer** water

**Capacitor** 25 µF

**Volume of DNA** less than 5 µl

**Resistor** 200 Ω (Pulse Controller)

## After the Pulse

**Time Constant** 5 msec

**Outgrowth Medium** AA/8 (see growth medium)

## Relevant Publications and/or Comments

**Note:** exponential values designated in parentheses.  
**Reference:** *J. Bacteriol.* **171**:5743-5746 (1989).

**Outgrowth Temperature** 30 °C

**Length of Incubation** 24 hours

**Selection Method or Assay Used** unspecified antibiotic / agar plates

**Electroporation Efficiency** varies with DNA concentration: 1x 10<sup>4</sup>(4) to 10<sup>6</sup>(6) transformants / µgDNA

**Per Cent Survival** 95%

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**Survey Number**

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